Analysis of the Mamba, Transformer, and CNN Architectures in the NIH ChestX-Ray14 Dataset

Introduction

Chest X-ray imaging is one of the most widely used tools for screening and diagnosing thoracic disease, but accurate automated interpretation remains difficult when a single image contains multiple coexisting abnormalities . The multi-label setting is especially challenging because thoracic findings often vary in size and location, can look visually similar, and frequently co-occur within the same radiograph. The NIH ChestX-ray14 dataset has become a standard benchmark for this problem because it contains over 112,000 frontal chest X-rays from more than 30,000 patients annotated for 14 thoracic diseases. However, the dataset also introduces major learning challenges, including severe class imbalance, label co-occurrence, and label noise from annotations automatically derived from radiology reports. Convolutional neural networks, especially DenseNet-based models, have been central to chest X-ray classification because they learn strong local image features and have historically achieved competitive performance on ChestX-ray14. DenseNet remains attractive in this setting because its densely connected design improves gradient flow, encourages feature reuse, and supports efficient representation learning in large-scale medical imaging tasks.

At the same time, CNNs are limited by their local receptive bias, which can reduce sensitivity to global context and long-range dependencies that matter in multi-label chest radiograph interpretation. Transformer-based studies improved this limitation by modeling broader image context and label relationships, and hybrid CNN–Transformer models often achieved better mean AUC than single-backbone baselines on ChestX-ray14. Recent Mamba-based state-space models offer another path to global modeling by capturing long-range dependencies with linear computational complexity rather than the quadratic scaling of attention. In chest X-ray research, Mamba variants have shown promising accuracy and substantially lower memory use, suggesting they are especially relevant when high-resolution medical images must be processed efficiently.

Abstract

This study proposes a deep learning framework for multi-label chest X-ray anomaly detection based on DenseNet, MedMamba, and a hybrid DenseNet–MedMamba architecture. The objective is to evaluate whether combining DenseNet's strength in local feature extraction with MedMamba's efficient long-range dependency modeling can improve classification of thoracic abnormalities. DenseNet-based models remain effective for medical image classification but are constrained by locality, whereas Mamba-based architectures offer linear-complexity global modeling with lower memory requirements. The proposed hybrid model integrates convolutional and Mamba-based representations to capture both fine-grained pathological textures and broader structural context. Prior hybrid studies across medical imaging show that combining CNN-style local encoding with Mamba modules and adaptive fusion can improve accuracy, efficiency, and robustness relative to single-backbone approaches. The project therefore benchmarks the standalone DenseNet and MedMamba models against the hybrid architecture to determine the most effective design for multi-label chest X-ray classification.